RA

RUTGERS ASPHALT PAVEMENT LABORATORY · CAIT

Rutgers Asphalt Analysis Tool-Pack

Version 3 — user instructions for the ten performance-test analysis modules, file templates, reports and master curves.

VERSION	3.0.0
REPLACES	RAAT-Pack v2 (MATLAB, 2018–2022)
PLATFORM	Windows 10/11 desktop application · no MATLAB Runtime, no Excel required
MODULES	APA · Beam Fatigue · Bond Strength · Cold SCB · DCT · Dynamic Modulus · Hamburg · HT-IDT · IDEAL-CT · Intermediate SCB
ORIGINAL AUTHORS	Thomas Bennert, Ed Haas, Benjamin Berger, Marissa Berger

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How to use this manual

Sections 1–10 apply to every module and are worth reading once. Section 11 has a two-page entry per test giving its input format, the dimensions it needs, what it reports and any step where the software asks you a question. If you have used Version 2, start with section 2.

The video instructions that shipped with Version 2 still describe the same tests and the same underlying calculations, but the screens they show are the old ones.

SECTION 1

About the Tool-Pack

Modern asphalt performance testing pairs simple physical procedures with intensive analysis of the collected data. The RAAT-Pack was created at the Rutgers Asphalt Pavement Laboratory to take the calculus and curve fitting off the technician's desk — reducing operator error and cutting the time needed to report a set of specimens, so that tests like the SCB can realistically be used for QC/QA.

The software supports ten tests:

MODULE	STANDARD	MEASURES
APA	AASHTO T340	Early-life rutting under a wheel-track
Beam Fatigue	AASHTO T321 · ASTM D7460	Fatigue life under repeated flexural bending
Bond Strength	AASHTO TP 114	Interlayer shear strength between pavement lifts
Cold Temperature SCB	AASHTO TP 105-13	Low-temperature fracture energy
DCT	ASTM D7313	Low-temperature fracture energy (compact tension)
Dynamic Modulus	AASHTO T342 · R62 · R84	Modulus master curves from AMPT data
Hamburg	AASHTO T324-17	Rutting and moisture susceptibility
HT-IDT	ASTM D6931 (modified)	High-temperature indirect tensile strength
IDEAL-CT	ASTM D8225	Cracking resistance (CT Index)
Intermediate Temperature SCB	AASHTO TP 124	Cracking resistance (Flexibility Index)

The calculations are unchanged. Version 3 is a re-implementation of the Version 2 analysis code, not a re-derivation. Every equation, constant and threshold was ported from the original MATLAB and checked against the summary reports Version 2 produced from the laboratory's demo specimens. Where Version 2 had a quirk that affected a reported number, that quirk was kept deliberately so your historical results stay comparable. Appendix B lists them.

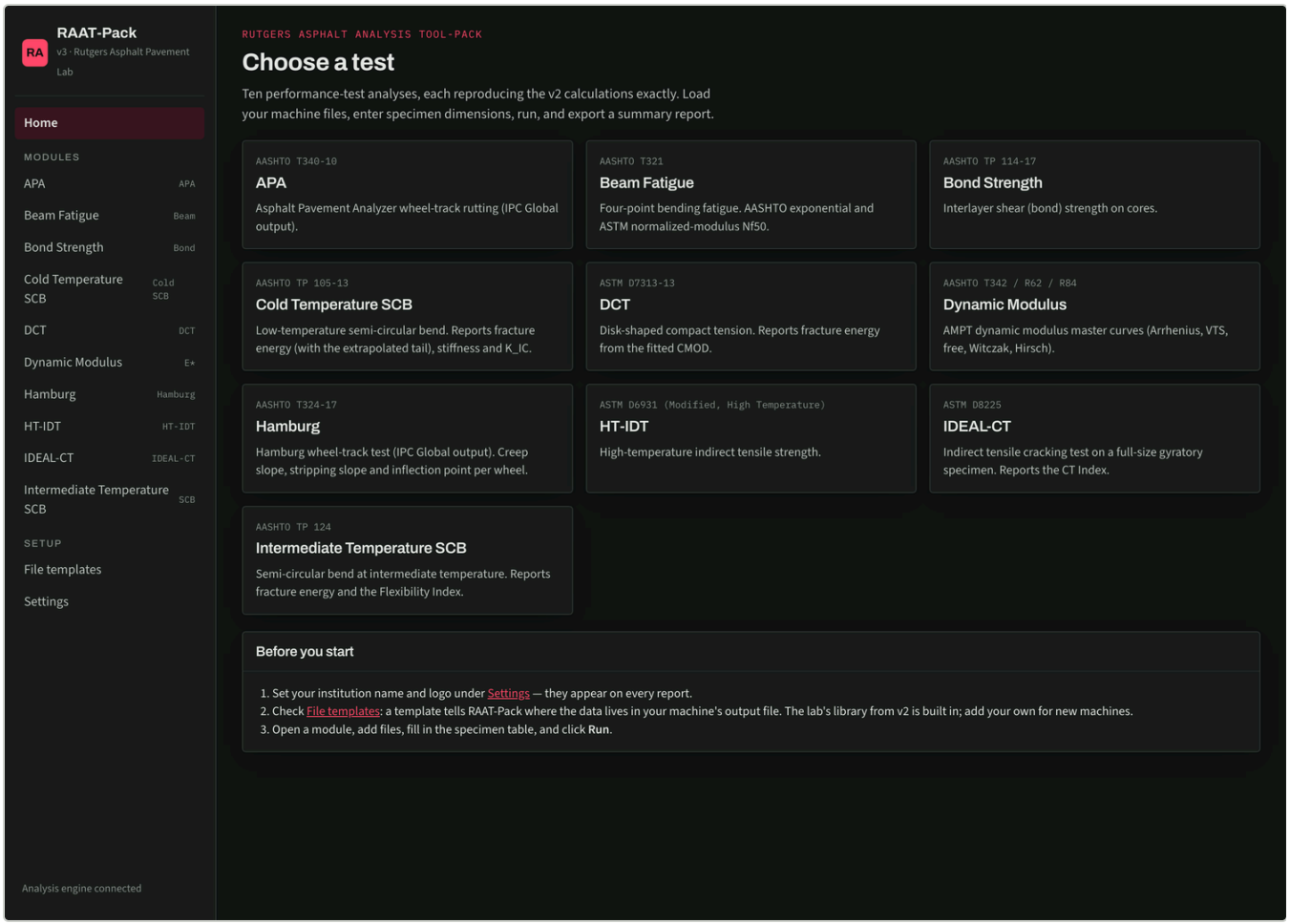


Figure 1.1 — *The home screen. Every module is reachable from the sidebar or from these cards.*

SECTION 2

What changed from Version 2

If you used Version 2, the analysis you get is the same. What changed is everything around it.

Things you no longer have to do

VERSION 2	VERSION 3
Install the MATLAB Runtime 9.4 (and only 9.4)	Nothing to install beyond the application itself
Microsoft Excel had to be installed, and the program drove it in the background to build every report	Reports are written directly. Excel is optional — it is only needed if you want to *edit* the workbook afterwards
Maximum of 12 specimens per summary report	No limit
A separate “RU” report layout behind a login	One report layout; your institution name and logo are set once in Settings
Each module opened its own window	One window; modules share the same five-step layout
Pop-up dialogs interrupted the analysis to ask about slopes and trims	The same questions are asked on the chart, where you can see what you are choosing, and can be changed or undone afterwards
Windows only	Windows for the packaged application; the analysis engine itself also runs on macOS and Linux

Things that are new

- **The construction is drawn on the curve.** You can see the 85 %–65 % slope window on an IDEAL-CT curve, the tangent at the SCB inflection point, the extrapolated Cold SCB tail, and the Hamburg creep and stripping lines — instead of only the numbers they produce.
- **Extrapolation is flagged.** Where Hamburg reports rutting at 20 000 passes for a test that stopped earlier, the chart shades the extrapolated region and the results table marks the row.
- **Statistics you can see.** The mean, standard deviation and COV that appear on the report are also shown as a dot plot per parameter, with the replicate that drives a high COV highlighted.
- **Data inspection.** `Inspect data points` shows the numbers the analysis actually used, after the template's unit conversions and sign corrections.
- **Sessions.** A run — including any manual picks — can be saved to a file and reopened later to reproduce the same numbers.
- **Template editor.** File templates can be created and checked against a sample file inside the application, rather than by editing a spreadsheet by hand.
- **More export formats.** Excel, PDF, CSV, and the chart on its own as a PNG.

> **Your Version 2 templates still work.** The whole template library from Version 2 is built in, and any additional template workbook you made can be imported unchanged (section 6). Nothing needs to be re-entered.

SECTION 3

System requirements & installation

Requirements

OPERATING SYSTEM	Windows 10 or Windows 11, 64-bit
DISK	≈ 400 MB for the application; results and uploads accumulate in your user folder
MEMORY	4 GB is sufficient for normal specimen counts
EXCEL	**Not required.** Only needed if you want to open and edit an exported workbook
MATLAB	**Not required.** No runtime, no licence
INTERNET	Not required. The application runs entirely on your machine

The window is drawn by the Microsoft Edge WebView2 runtime, which is part of Windows 10 and 11. On a machine where it has been removed, Windows Update or the free WebView2 Evergreen installer restores it.

Installing

1. Copy the `RAATPack` folder to the machine — for example to `C:\RAATPack` or to the desktop. Keep the folder together; the application will not run if `RAATPack.exe` is moved out of it on its own.
2. Double-click `RAATPack.exe`. The first launch takes a few seconds longer while Windows inspects the files.
3. Optionally, right-click `RAATPack.exe` → **Send to** → **Desktop (create shortcut)**.

There is no installer to run and nothing is written to the Windows registry. To remove the software, delete the folder.

Where your data is kept

Settings, custom templates, the raw files you load and every report you export are stored under your Windows user profile:

```
C:\Users\<you>\AppData\Roaming\RAATPack\
    settings.json      institution, logo, preferences
    templates\         file templates you created
    uploads\           copies of the machine files you loaded
    exports\           every report you have exported
```

The exact path is shown at the bottom of the Settings screen. Exports are also offered as a normal browser download when you press Export, so you can save a copy wherever you like; the copy in `exports\` is a safety net.

> **Where to get it.** The packaged application is published on the project's GitHub Releases page — `github.com/ShaaravJag/RAATPack/releases`. Download the `RAATPack-…-windows.zip` from the most recent release, unzip it, and follow the steps above. No GitHub account is needed. Each release is built and launch-tested automatically on Windows before it is published.

> **Running without the packaged application.** The same software can be run from the source folder on Windows, macOS or Linux — see Appendix C. Everything in this manual applies either way; only the way you start it differs.

SECTION 4

First-time setup

Two things are worth setting before you analyse anything: who the reports come from, and which file templates match your machines.

Settings

Open **Settings** at the bottom of the sidebar.

INSTITUTION	Printed in the header of every report. Pre-filled with “Rutgers Asphalt Pavement Laboratory”.
DEFAULT TECHNICIAN	Pre-fills the Technician box on each module. You can still change it per project.
LOGO	A PNG or JPG placed at the top-left of every report. In Version 2 this had to be set before the program would let you export; here it is optional.
DEFAULT EXPORT FORMAT	Which format the Export panel offers first.
COV WARNING THRESHOLD	Coefficients of variation above this are highlighted amber in the results table and statistics panel. It is a visual aid only — it never changes a reported number, and it is not written to the report. Default 15 %.
THEME	Light, dark, or follow the operating system.

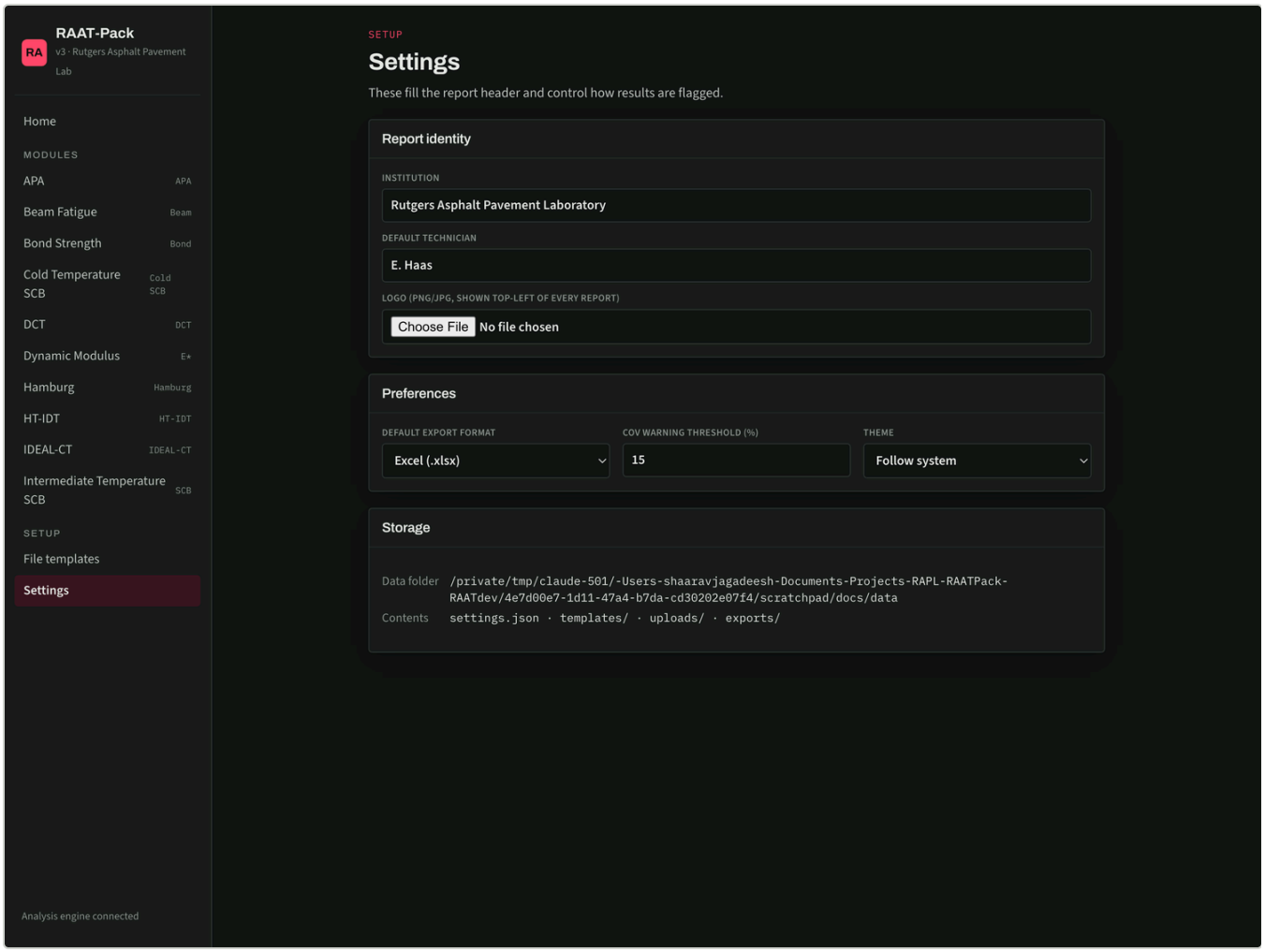


Figure 4.1 — *Settings. The data folder path at the bottom is where exports and custom templates are kept.*

Templates

A *file template* tells the software where the numbers are inside your machine's output file. You will normally pick one from the built-in list and never think about it again. Section 6 explains how to add one for a machine or export format that is not covered.

SECTION 5

How every module works

All ten modules use the same five-step page. Learning one teaches you the rest; only the dimensions you type and the values you get back change.

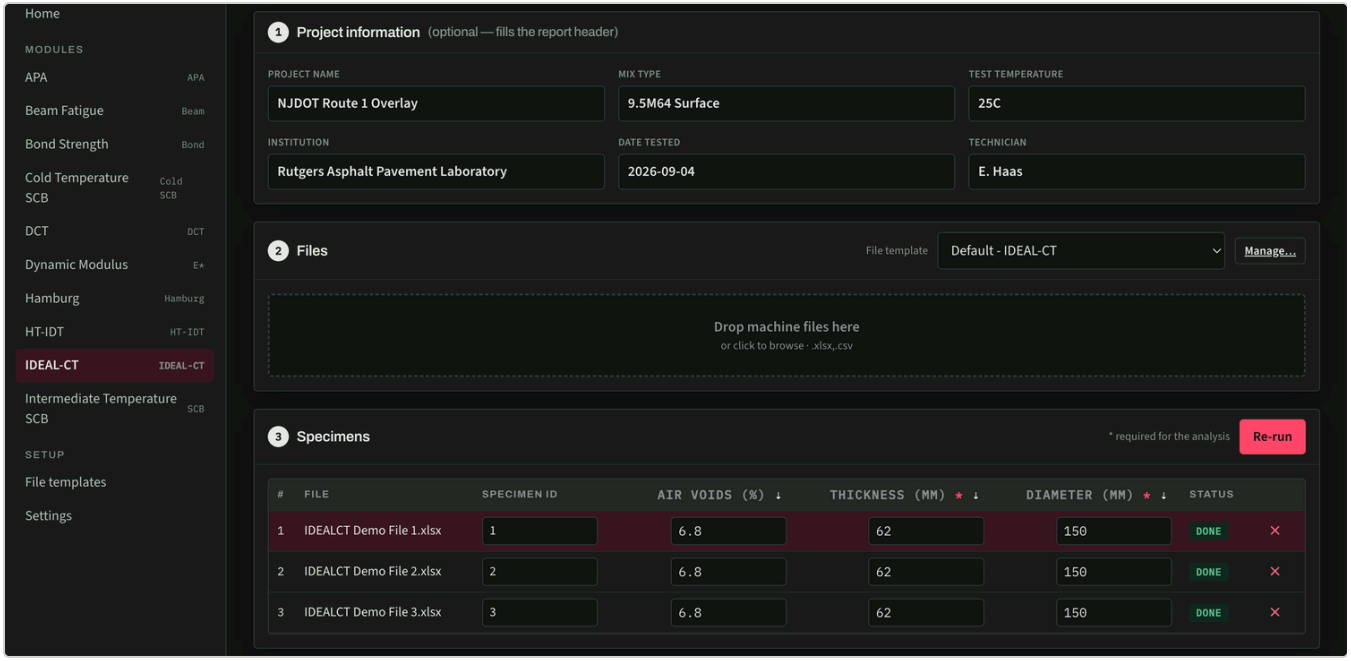


Figure 5.1 — *Steps 1–3, shown on IDEAL-CT. The same three panels appear on every module.*

Step 1 — Project information

Project name, mix type, test temperature, institution, date and technician. All optional; they populate the report header. Institution and technician are pre-filled from Settings. Values are remembered while the application is open, so a second module in the same project starts blank only for the fields you have not filled.

Step 2 — Files

Choose the file template that matches your machine (where the module needs one), then drag your files onto the drop zone or click it to browse. Several files can be added at once, and you can add more later without losing what is already loaded.

Accepted formats are `.xlsx` and `.csv`. If a file cannot be read, it still appears in the specimen table with a red **error** badge — hover it for the reason. The usual cause is the wrong template.

> **APA, Hamburg and Dynamic Modulus need no template.** They read fixed manufacturer formats — the IPC Global workbook for the two wheel-track tests, and the AMPT `…Sum.csv` files for Dynamic Modulus.

Step 3 — Specimens

One row per file. Give each specimen an ID (if you leave it blank it becomes “Sample #1”, “Sample #2”…), then enter the dimensions. **Columns marked with a red asterisk are required** — the analysis cannot run without them, and empty required boxes are outlined in amber.

The small ⇣ button in a column header copies the first row's value down the whole column — useful when a set of specimens shares a nominal thickness or diameter.

Press **Run analysis**. Each file is analysed independently: one bad specimen produces an error badge on its own row and does not stop the others.

Step 4 — Results

The table gives one row per specimen plus Average, Std Dev and COV — the same three statistics Version 2 printed. Clicking a row selects that specimen for the chart. COV values above your warning threshold are marked amber.

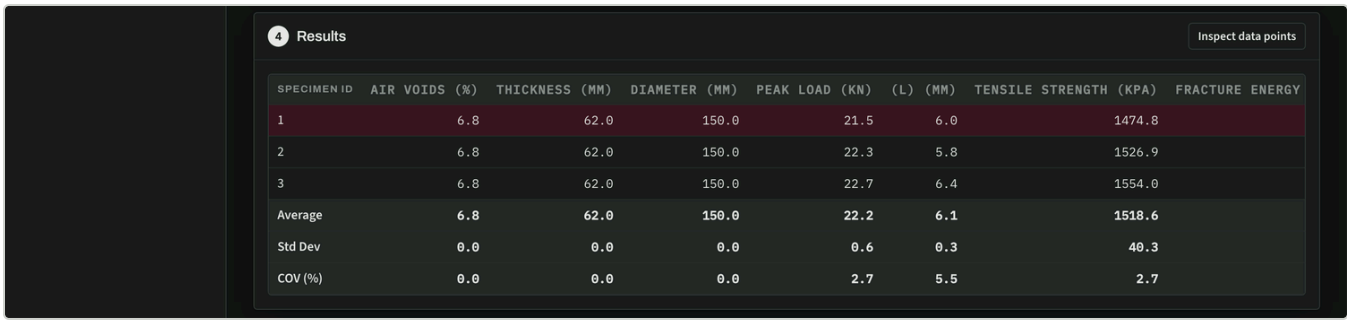


Figure 5.2 *— Results for three IDEAL-CT specimens. The amber COV on $Gf/S{\cdot}(L/D)^2$ is above the 15 % threshold.*

Inspect data points opens the actual numbers used by the analysis — after the template's unit conversion and sign corrections have been applied. This is the quickest way to confirm a template is doing what you expect.

Step 5 — Export

Covered in section 9.

SECTION 6

File templates

Every make of testing machine writes its results differently — different columns, different header rows, different units, sometimes with the load recorded as a negative number. A template captures those differences once so that the analysis itself never has to care.

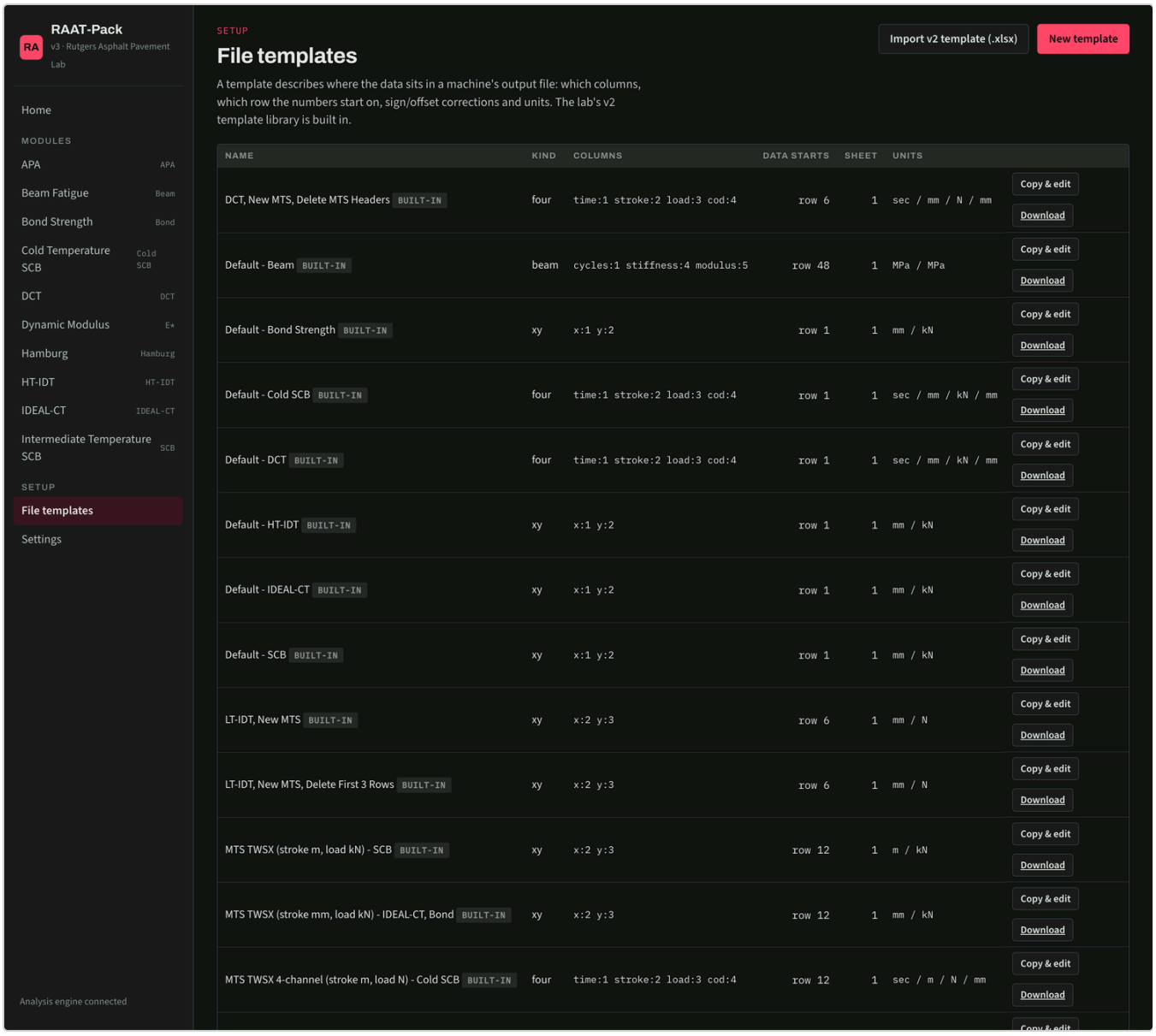


Figure 6.1 — *The template library. Built-in templates are marked; the laboratory's Version 2 templates are included.*

What is in the list

- **Default – …** the per-module defaults from Version 2. These expect a file already formatted for the RAAT-Pack: data starting in row 1, displacement in column 1, load in column 2.

- **MTS TWSX …** the raw export from the laboratory's MTS frames, including the header rows and the sign conventions, so no pre-formatting is needed.
- **Pine …, SmartJIG …, New MTS …, LT-IDT …, SCB, New MTS …** the templates the laboratory built up during Version 2, carried over unchanged.

> **File Input Examples.xlsx**, in this same Instructions folder, shows exactly what each **Default – …** template expects, one sheet per test. It is still accurate for Version 3. You only need it if you are formatting data by hand rather than pointing a template at the machine's own output.

Choosing the right one

Pick the template whose name matches the machine and the test. If you are unsure, load one file, run it, and use `Inspect data points`: displacement should start near zero and increase, load should be positive and rise to a peak. If either is negative or reversed, the template is wrong.

Creating a template

1. Press `New template` and give it a name that identifies the machine and the test, e.g. “Pine SCB (CSV)”.
2. Choose the kind. **Two-channel** (displacement and load) covers IDEAL-CT, SCB, HT-IDT and Bond Strength. **Four-channel** (time, stroke, load, COD gauge) covers Cold SCB and DCT. **Beam fatigue** covers cycles, flexural stiffness and modulus.
3. Enter the column numbers, the row the numbers start on, the sheet number for workbooks, and the units of each channel.
4. Set the corrections — see below.
5. Load a sample file on the right. The first rows of the file are shown with the row numbers the template sees, greyed above your data-start row, and the parsed result is summarised. Adjust until it reads correctly, then `Save template`.

Corrections

Each channel has three flags:

OFFSET FROM START	Subtract the first reading, so the channel begins at zero
NEGATIVE	Flip the sign — for machines that record compression as negative load
OFFSET FROM END	Subtract the last reading

They combine: “offset from start” together with “negative” gives $-(x - x_1)$, which is what turns a decreasing stroke reading into an increasing displacement.

Importing and sharing

`Import v2 template (.xlsx)` reads a template workbook from Version 2 exactly as it was. `Download` writes any template back out in the same workbook format, so templates can be emailed between machines or laboratories.

Built-in templates cannot be overwritten. `Copy & edit` makes an editable copy, leaving the original intact.

SECTION 7

Reading the charts

The chart panel has two views. **All specimens** overlays the set so you can compare replicates. **Selected + construction** shows one specimen with the geometry the analysis derived from it drawn on top.

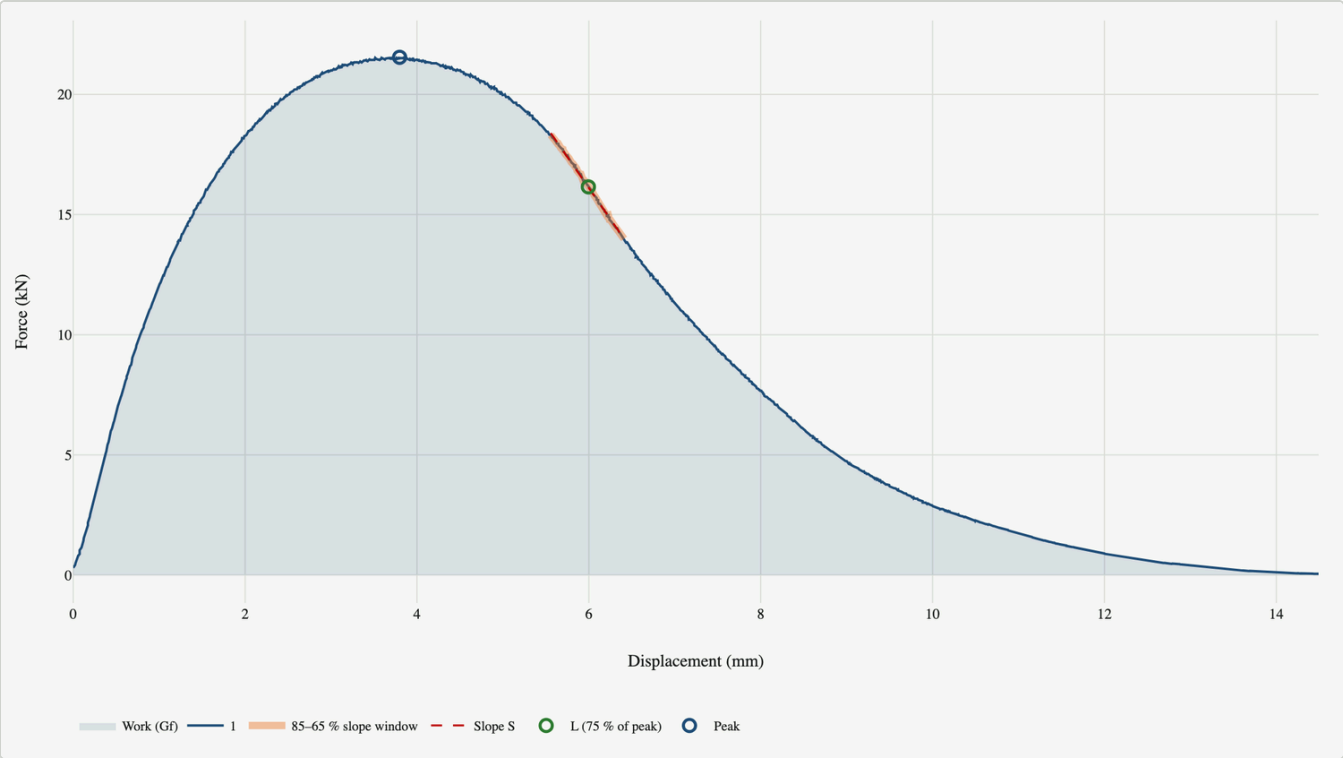


Figure 7.1 — *IDEAL-CT construction: the shaded area is the work used for fracture energy; the orange band is the 85 %–65 % post-peak window the slope S is fitted over; the dashed red line is that slope; the green marker is L, the displacement at 75 % of peak load.*

Every element in the legend can be switched off by clicking its pill above the chart, which is useful when overlays hide the data. The chart is interactive: drag to zoom, double-click to reset, and use the camera icon in the top-right to save the current view as an image.

What each module draws

MODULE	CONSTRUCTION SHOWN
IDEAL-CT	Peak, 85 %–65 % slope window, fitted slope, L at 75 % of peak, shaded work area
Intermediate SCB	Post-peak 6th-order fit, tangent at the inflection point, the inflection point, other candidate roots, shaded work area
Cold SCB	Pre-peak stiffness line, the 60 %-of-peak point where the tail fit starts, the extrapolated tail (dashed), shaded W and W_{tail}; plus COD gauge against time
DCT	Peak, shaded work area; plus measured COD against the fitted CMOD used for the integral
Hamburg	Both wheels, creep and stripping slope lines, the stripping inflection point, and a shaded band where the 20 000-pass rut is extrapolated
Beam Fatigue	Flexural stiffness against cycles on log axes with the AASHTO exponential fit; plus the ASTM normalized-modulus curve with its 6th-order fit and $n_{f,50}$
APA	Average rutting against cycles, with each active wheel shown separately
HT-IDT, Bond Strength	Load–displacement curve with the peak marked

SECTION 8

Statistics

The statistics panel presents the same average, standard deviation and coefficient of variation that appear in the results table and on the exported report. Nothing new is calculated; the numbers are simply easier to judge.

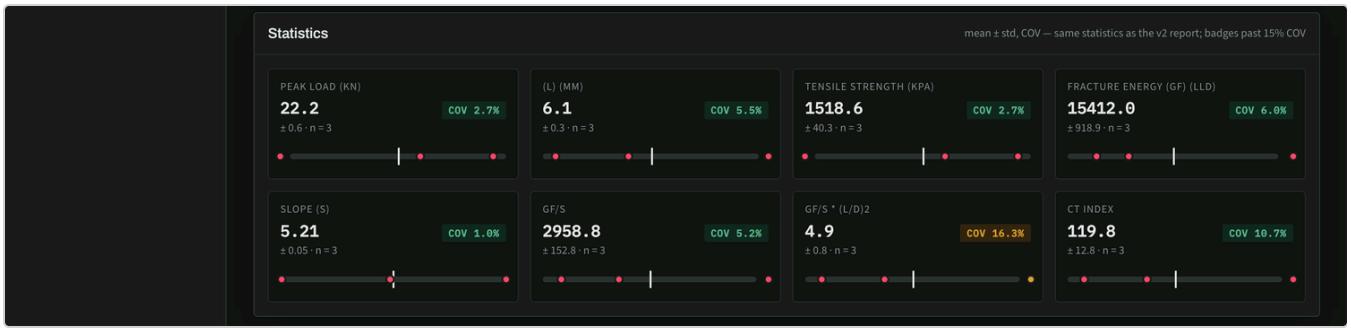


Figure 8.1 — *One tile per reported parameter. The vertical rule is the mean, the grey band is ± one standard deviation, and each dot is a replicate.*

Hovering a dot names the specimen it belongs to. When the COV is above your warning threshold the badge turns amber and the replicate furthest from the mean is coloured, which points straight at the specimen driving the scatter — usually the one to re-check for a mis-typed dimension or a bad trim.

> **These are presentation aids only.** The COV threshold does not appear on the report and does not mark a specimen as failing. Acceptance remains a matter for the test method and the specification you are working to.

SECTION 9

Reports and exports



Figure 9.1 — *The export panel. Specimen chips select what goes into the report.*

Formats

EXCEL (.XLSX)	The summary sheet, laid out as in Version 2, with a live chart and a *Sample Data* sheet holding every curve. Average, Std Dev and COV are written as formulas, so the sheet stays correct if you edit a value.
PDF	The same report as a fixed page, ready to sign or attach.
CSV	The results table only — for pasting into a spreadsheet of your own.
PNG	The chart on its own at presentation resolution.

Choosing specimens

The chips at the bottom of the panel select which analysed specimens go into the report; click to toggle. There is no limit — Version 2's twelve-specimen ceiling is gone.

> **APA, Hamburg and Beam Fatigue report one specimen at a time.** Their report layouts are built around a single test, exactly as in Version 2. Selecting a second chip replaces the first.

Per-module options

IDEAL-CT	A pass/fail criterion can be entered; it is printed under the table.
APA	Hose pressure and wheel load are printed in the test-conditions line.
COLD SCB	Whether to include the interpolated tails — all, none, or only the specimens you choose. This is the same question Version 2 asked at export time.

Where reports go

Pressing Export both downloads the file and keeps a copy in the application's `exports` folder. **Open exports folder** opens that folder in Windows Explorer.

SECTION 10

Saving and reloading a session

A session is everything you set up for a module: the files, the specimen IDs and dimensions, the project information, and any manual choices you made on the charts.

- **Save session** writes a small `.json` file. Keep it beside the raw data for the project.
- **Load session** reads it back, re-opens the same files and restores everything. Press **Run** and you will get exactly the numbers you had before, including any inflection point or slope you picked by hand.
- **Clear** empties the module and starts again.

> **A session refers to the original files by their location.** If the raw data is moved or deleted, the session will report that it cannot re-open them. Save the session next to the data, not somewhere else.

This matters most for Intermediate SCB and Hamburg, where a manual pick affects the reported value: the session is the record of what was chosen, so a result can be reproduced or reviewed months later.

SECTION 11

Modules

One entry per test, in the order they appear in the sidebar. Each gives the input format, the dimensions required, what is reported, and any point where the software asks you a question.

Throughout, dimensions marked **required** must be filled before `Run analysis` will produce a result for that specimen. Air voids and sample ID are always optional and are carried through to the report for identification only.

MODULE	TEMPLATE NEEDED	REQUIRED DIMENSIONS	SPECIMENS PER REPORT
APA	no	none	one
Beam Fatigue	yes — beam	none	one
Bond Strength	yes — two-channel	length, diameter	many
Cold Temperature SCB	yes — four-channel	thickness, radius, notch	many
DCT	yes — four-channel	thickness, ligament	many
Dynamic Modulus	no	volumetrics — see 11.6	one curve
Hamburg	no	none	one
HT-IDT	yes — two-channel	thickness, diameter	many
IDEAL-CT	yes — two-channel	thickness, diameter	many
Intermediate SCB	yes — two-channel	thickness, ligament	many

11.1 APA

AASHTO T340-10

The Asphalt Pavement Analyzer runs a steel wheel over a rubber hose to simulate traffic loading and measure early-life rutting. One to three wheel positions may be in use.

INPUT	Standard IPC Global workbook. The analysis reads the second sheet (`RawData`).
TEMPLATE	None — the format is fixed.
DIMENSIONS	Air voids (optional, for reporting).
REPORTS	Left, middle and right rut depth (mm), their average, standard deviation and COV.
PER REPORT	One specimen.

1. Enter project information.
2. Add your workbooks. The software detects which wheel positions were used by looking for a non-zero column; unused positions are left blank rather than counted as zero.
3. Enter the air voids for each specimen if you want them on the report.
4. **Run analysis**. The rut depth for each wheel is its final reading.
5. Under Export, enter the hose pressure and wheel load — they are printed in the test-conditions line — then export.

The exported report includes the two NJDOT criteria lines carried over from Version 2: intermediate course (PG64-22) ≤ 7 mm and surface course (PG76-22) ≤ 4 mm.

If the file names do not appear after adding, the workbook is not in the IPC Global layout — check that the raw data really is on the second sheet.

11.2 Beam Fatigue

AASHTO T321 · ASTM D7460

A sinusoidal load applied to two inner clamps bends a cut asphalt beam against two outer reaction clamps. The analysis finds where the specimen reaches half its initial stiffness — its $n_{f,50}$. Both the AASHTO and the ASTM method are run on every file.

INPUT	The `.csv` written by the beam fatigue machine.
TEMPLATE	Yes. With **Default – Beam**, cycles are in column 1, flexural stiffness (MPa) in column 4 and elastic modulus (MPa) in column 5, with data from row 48 — that is, from cycle 50.
DIMENSIONS	Air voids, width, height, length — all optional, all for reporting only. The analysis uses the machine's stiffness and modulus columns directly.
REPORTS	AASHTO $n_{f,50}$ and ASTM $n_{f,50}$ in the table; the AASHTO trendline coefficients *A* and *b*, the stiffness and modulus at $n_{f,50}$, the six ASTM polynomial coefficients and the cycle at maximum normalized modulus are printed beneath it.
PER REPORT	One specimen.

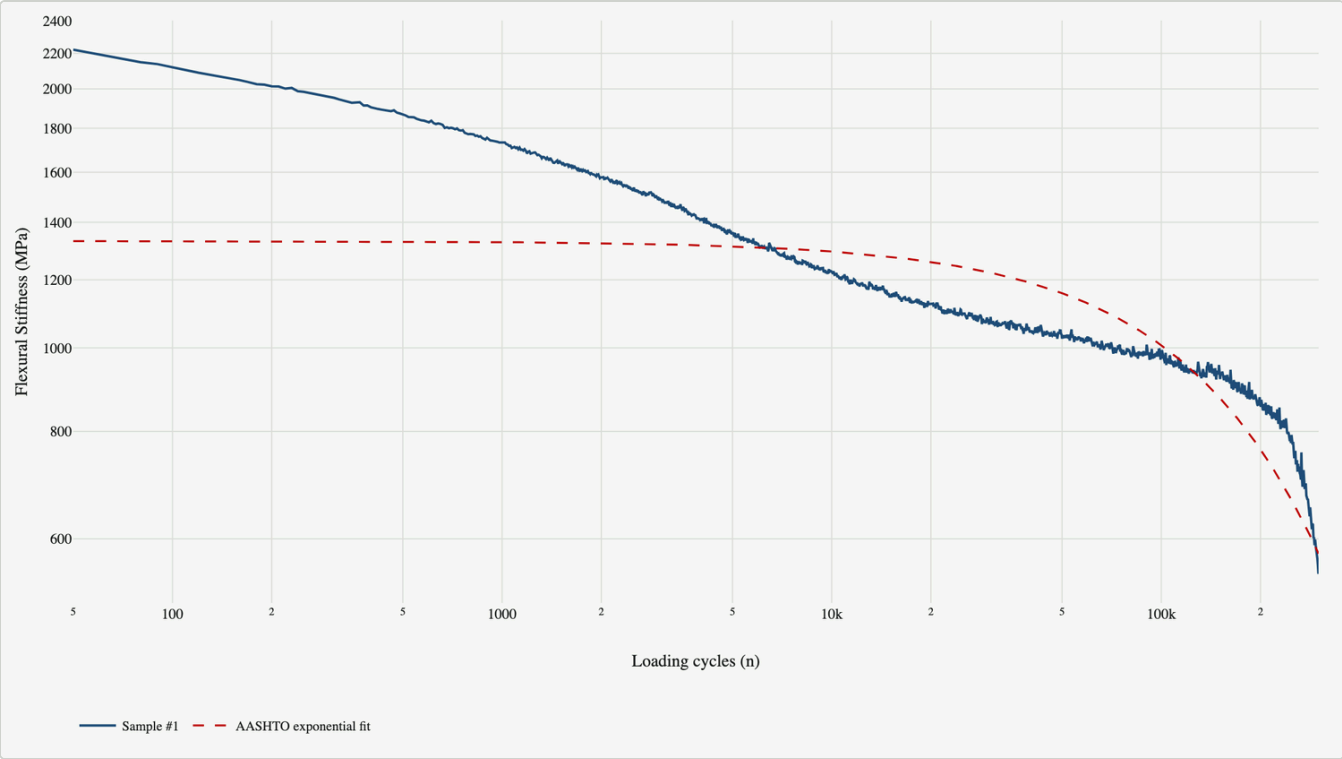


Figure 11.2a — *AASHTO method: flexural stiffness against cycles on log axes, with the robust exponential fit.*

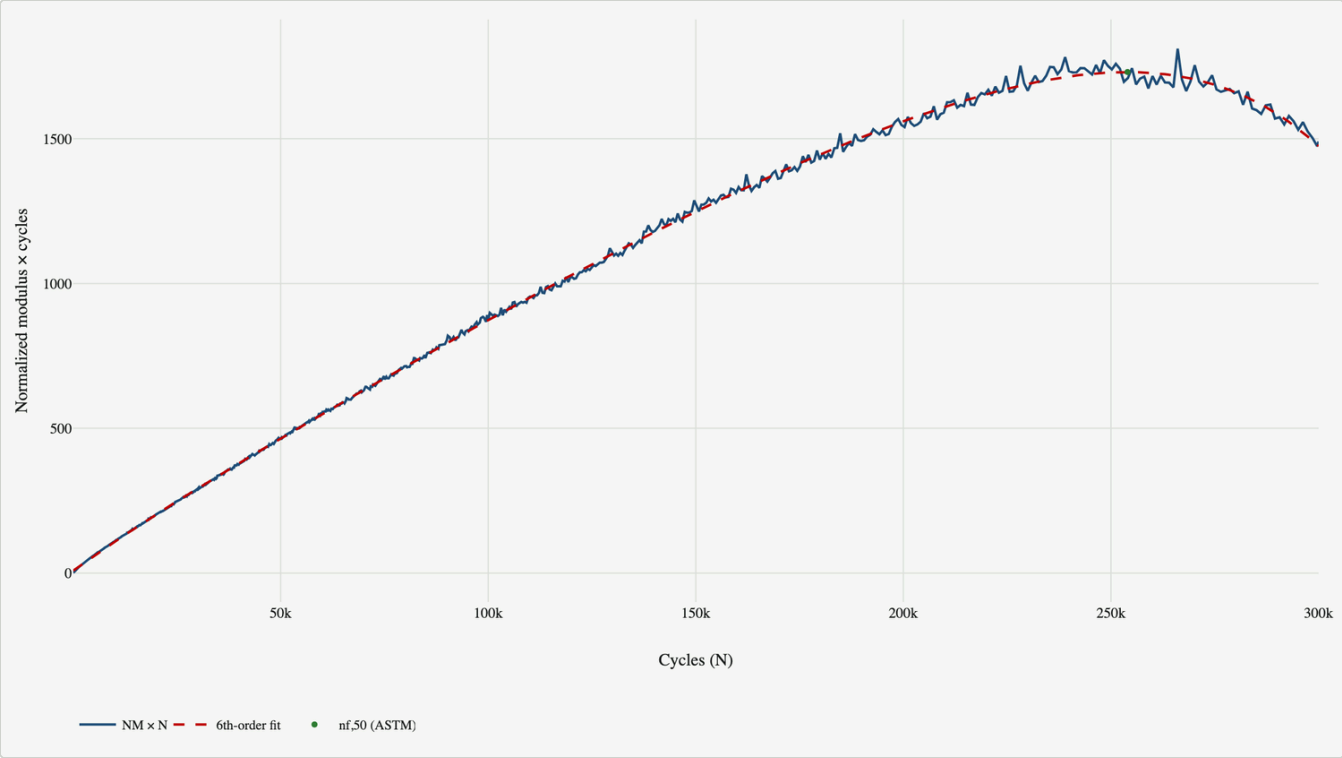


Figure 11.2b — *ASTM method: normalized modulus × cycles, its 6th-order fit, and $n_{f,50}$ at the peak.*

> If the file begins at cycle 10, the first four rows are skipped so that the analysis starts at cycle 50, as the test method requires. This matches Version 2.

11.3 Bond Strength

AASHTO TP 114-17

Interlayer shear strength characterises the adhesion between two pavement lifts. It is normally run on six-inch cores taken from in-service pavement; low results often point to poorly placed tack coat.

INPUT	Two columns: stroke and load.
TEMPLATE	Yes — two-channel. **Default – Bond Strength** expects positive stroke in mm starting at zero in column 1 and positive increasing load in kN in column 2. For the raw MTS export use **MTS TWSX (stroke mm, load kN)**.
DIMENSIONS	**Length (mm)**, **diameter (mm)**; air voids optional.
REPORTS	Length and diameter converted to inches, peak load (kN), displacement at peak load (mm), bond strength (psi).
PER REPORT	Many specimens.

1. Enter project information and choose the template.
2. Add your files.
3. Enter length and diameter for each core.
4. **Run analysis**, then export.

11.4 Cold Temperature SCB

AASHTO TP 105-13

A semi-circular specimen is loaded to failure at low temperature. Fracture energy is obtained from the work done, plus an extrapolated “tail” that accounts for the load the specimen would continue to carry beyond the end of the record.

INPUT	Four columns: time (s), stroke, load, COD gauge.
TEMPLATE	Yes — four-channel. **Default – Cold SCB** expects positive time from zero, positive stroke in mm from zero, positive load in kN, positive COD in mm. For the raw MTS export use **MTS TWSX 4-channel (stroke m, load N)**.
DIMENSIONS	**Thickness**, **radius**, **notch length**, all in mm; air voids optional.
REPORTS	Peak load, time at peak, U_c, stiffness (kN/m), K_{IC} (MPa·m$^{0.5}$), fracture energy (J/m²) and CMOD slope. W, W_{tail}, W_{total} and the fracture energy with and without the tail are shown on screen for checking.
PER REPORT	Many specimens.

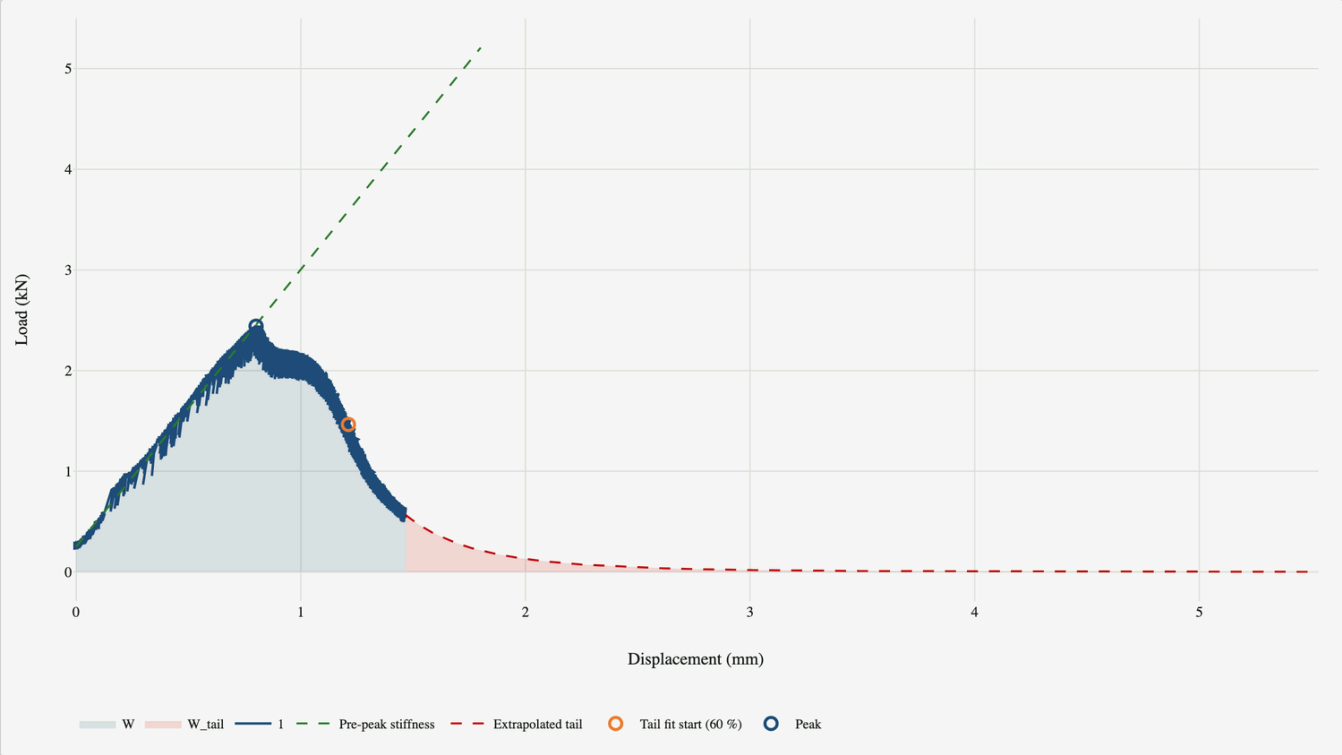


Figure 11.4 — *The pre-peak stiffness line, the 60 %-of-peak point where the tail fit begins, and the extrapolated tail continuing past the measured record.*

The tail

A power-law curve is fitted to the post-peak record from 60 % of peak load onward, then extended in 0.001 mm steps until the load falls below 0.001 kN. The area under that extension is W_{tail}.

If the fit will not converge to a decaying curve within 10 000 steps, the specimen is flagged `no tail` and its fracture energy is reported without one. Version 2 raised a warning dialog at this point; here the row is simply marked and a note appears under the table.

At export you are asked what to do with the tails — include them all, remove them all, or include only the specimens you select. This is the same choice Version 2 offered.

11.5 DCT

ASTM D7313-13

Disk-shaped compact tension. A circular specimen with a single edge notch is loaded in tension at low temperature to obtain fracture energy.

INPUT	Four columns: time (s), stroke, load, COD gauge.
TEMPLATE	Yes — four-channel. For the raw MTS export use **MTS TWSX 4-channel (stroke mm, load kN)**.
DIMENSIONS	**Thickness** and **ligament length** in mm are required; diameter and air voids are for reporting.
REPORTS	Peak load, displacement at peak, time at peak, fracture energy (J/m^2) and CMOD slope (mm/s). Test duration is shown on screen.
PER REPORT	Many specimens.

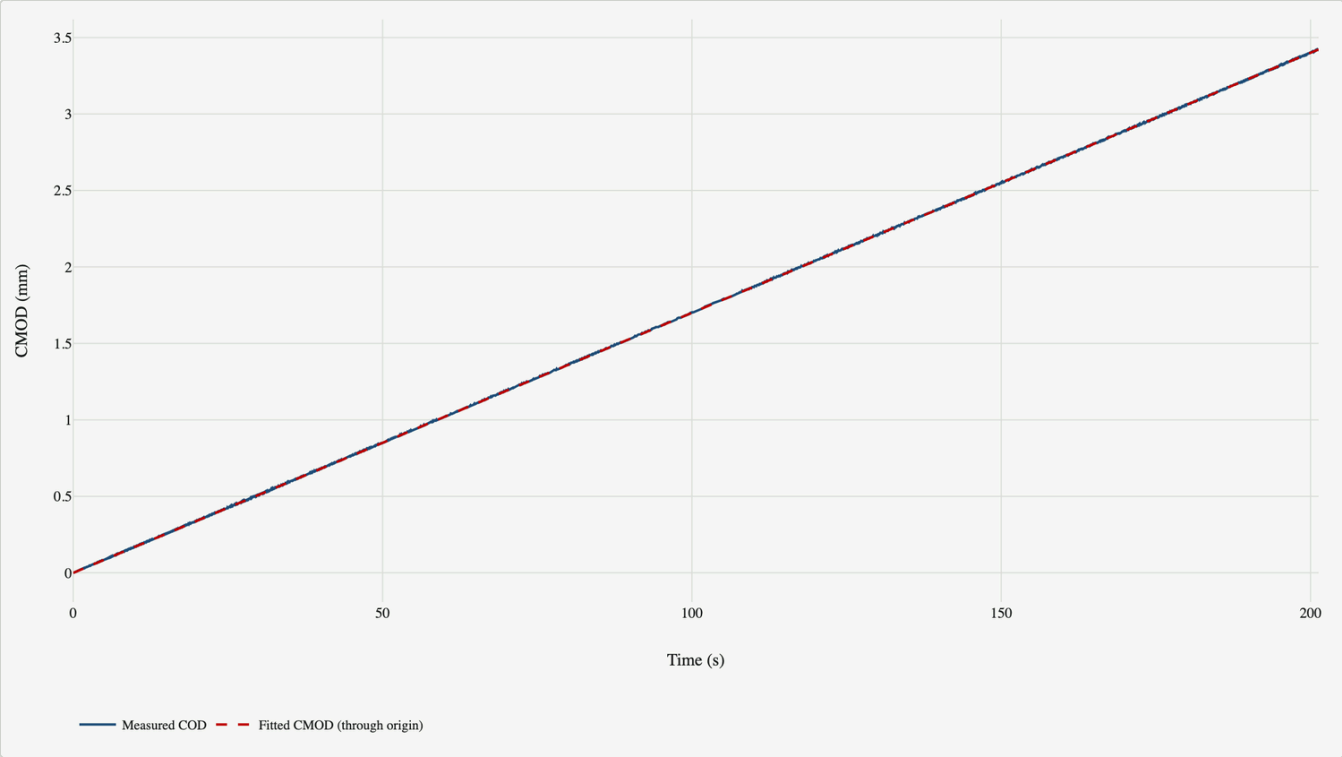


Figure 11.5 — *The CMOD rate check. The dashed line is the straight-line CMOD fitted through the origin; the fracture energy integral is taken against this fitted opening, not the raw gauge trace.*

> Displacement at peak load is read from the fitted CMOD rather than the stroke. This was corrected during Version 2 and is preserved here. Use the second chart to confirm the gauge opened at a steady rate — a curved trace means the test itself is suspect.

11.6 Dynamic Modulus

AASHTO T342 · R62 · R84

The Asphalt Mixture Performance Tester measures modulus at several temperatures and loading frequencies. A master curve is built by shifting the individual isotherms onto a single reduced-frequency axis. This module has its own workflow rather than the standard five steps.

Step 1 — Volumetrics

Enter G_{mm}, G_{sb}, air voids and asphalt content, then **Calculate E*max**. The Hirsch model gives VMA, VFA and E^*_{max}, which every master curve needs as its upper asymptote.

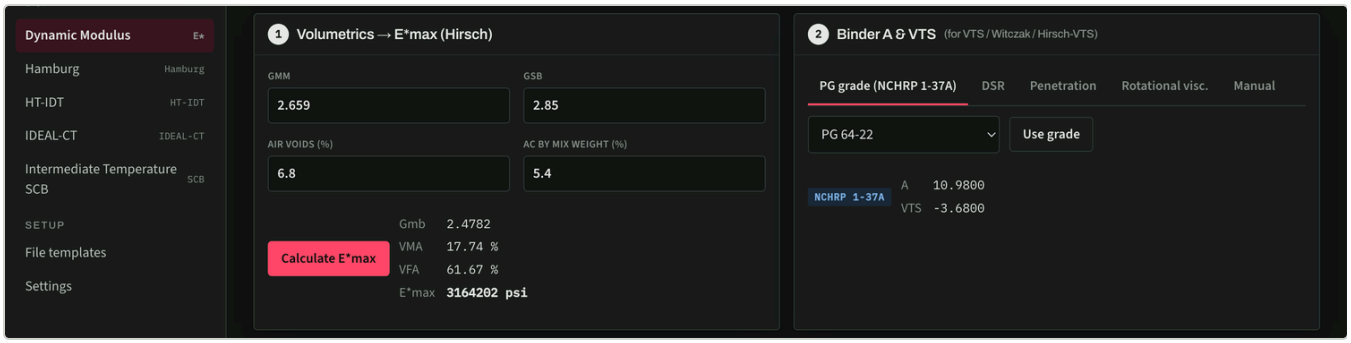


Figure 11.6a — *Volumetrics (left) and binder A & VTS (right).*

Step 2 — Binder A & VTS

Needed for VTS shifting, the Witczak equation and Hirsch-VTS; not needed for Arrhenius (EA) shifting. Choose one of:

- **PG grade** — the NCHRP 1-37A lookup table.
- **DSR** — temperature (°C), G* (Pa) and phase angle.
- **Penetration** — temperature (°F) and penetration.
- **Rotational viscometer** — temperature (°F) and viscosity (cP).
- **Manual** — type A and VTS directly.

For the three measured options the fit is plotted with its R^2, so you can see whether the data really is linear in log-log viscosity space.

Step 3 — AMPT files

Drop in the `…Sum.csv` files the AMPT software writes — one per specimen per temperature. Then assign them to a grid of up to three replicates × three temperatures and give each column its target temperature. **Auto-assign from file names** fills the grid using the replicate number and temperature in each file name.

Press **Load project files**. Frequencies are matched across replicates and averaged; where a file is missing a frequency the others have, that point is dropped rather than skewed.

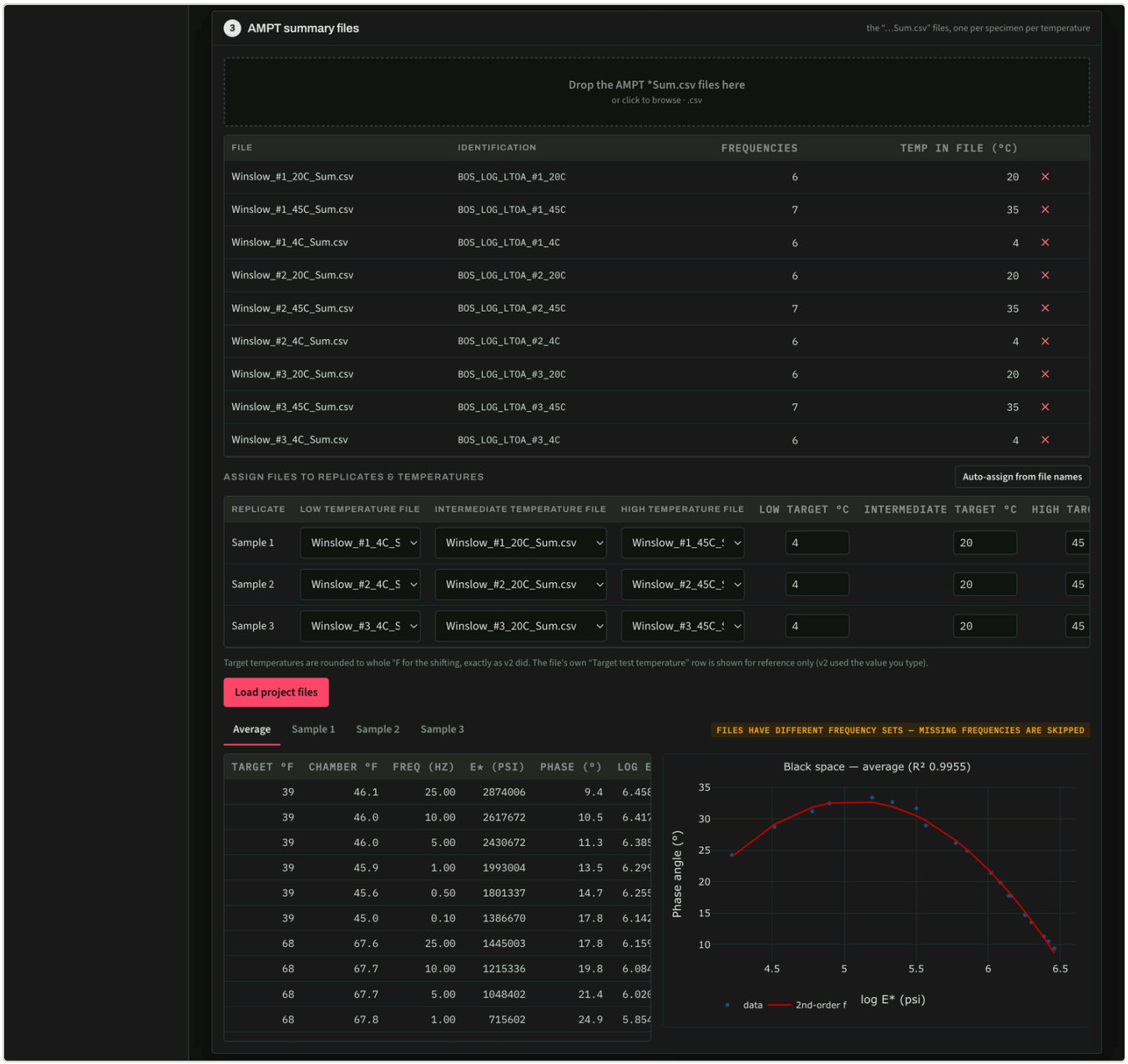


Figure 11.6b — *File assignment, the averaged data table with per-point COV, and the black-space plot.*

You can review the average or any individual replicate using the tabs, and the black-space plot (phase angle against log E*) gives a quick check on data quality — a tight, smooth arc means consistent data.

Step 4 — Witczak and Hirsch (optional)

Enter G_b and the gradation to run the Witczak 1-37A prediction, or load a workbook of binder G* against temperature and frequency for the Hirsch model. Neither is needed for the standard AMPT master curve.

Step 5 — Master curves

Tick the curves you want, set the reference temperature, choose whether to shift on the target temperature or the measured chamber temperature, and press **Run master curves**.

CURVE	NEEDS
AMPT (EA shifting)	AMPT data + E*max
VTS shifting	+ binder A & VTS
No-constraints shifting	AMPT data + E*max
Witczak prediction equation	+ binder, volumetrics, gradation
Hirsch model — EA / VTS shifting	+ the Hirsch G* file

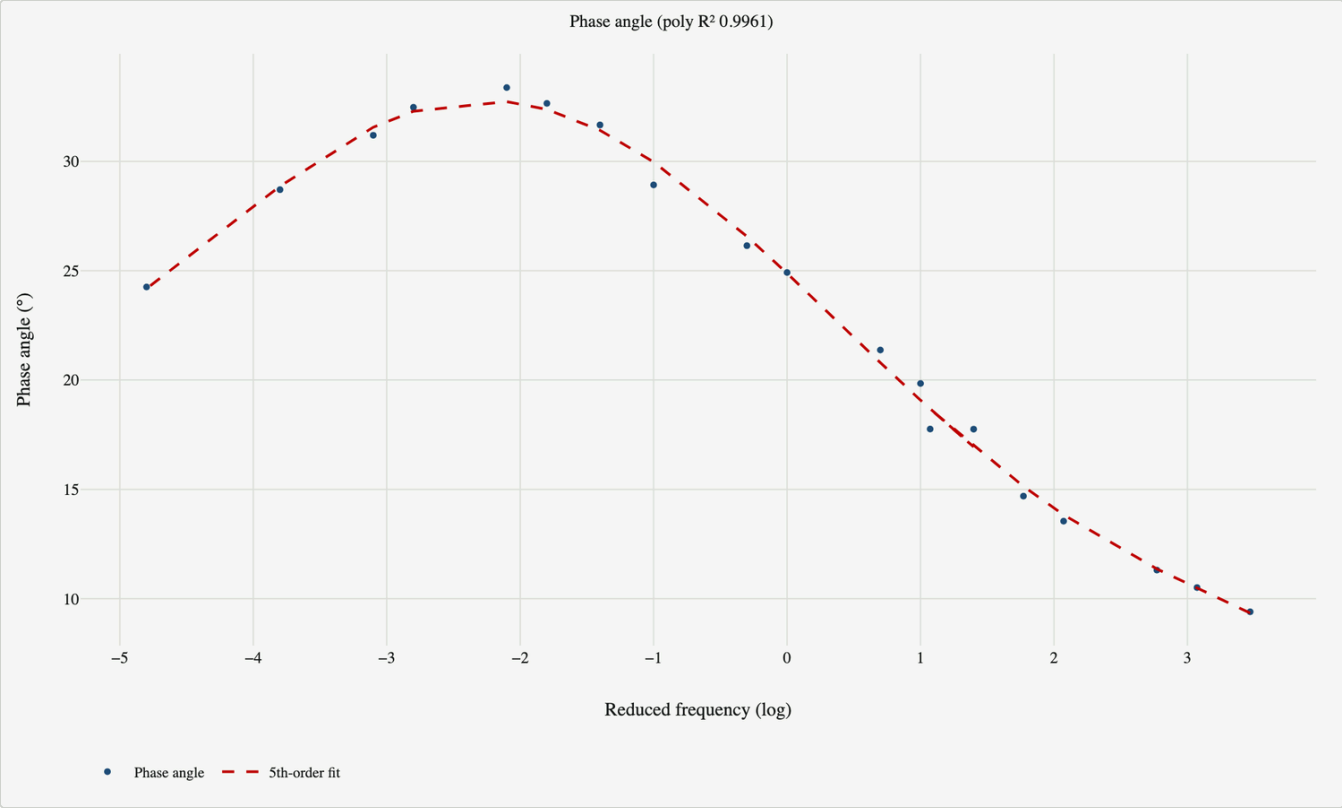


Figure 11.6c — *A fitted master curve. Measured points are shifted onto the reduced-frequency axis; the line is the fitted sigmoid.*

Each fitted curve shows its sigmoid coefficients, the sum of squared error, and the shift factor applied to each temperature. Export writes the master curve, the measured and predicted values and the phase angles to a two-page report.

11.7 Hamburg

AASHTO T324-17

Steel wheels pass over a submerged specimen to induce rutting. The specimen creeps, and may strip if the binder debonds from the aggregate. Both rut depth and moisture susceptibility come from the same record.

INPUT	Standard IPC Global workbook, second sheet.
TEMPLATE	None.
DIMENSIONS	Air voids (optional).
REPORTS	Per wheel and averaged: rutting at 20 000 cycles, creep slope, stripping slope and the stripping inflection point.
PER REPORT	One specimen.

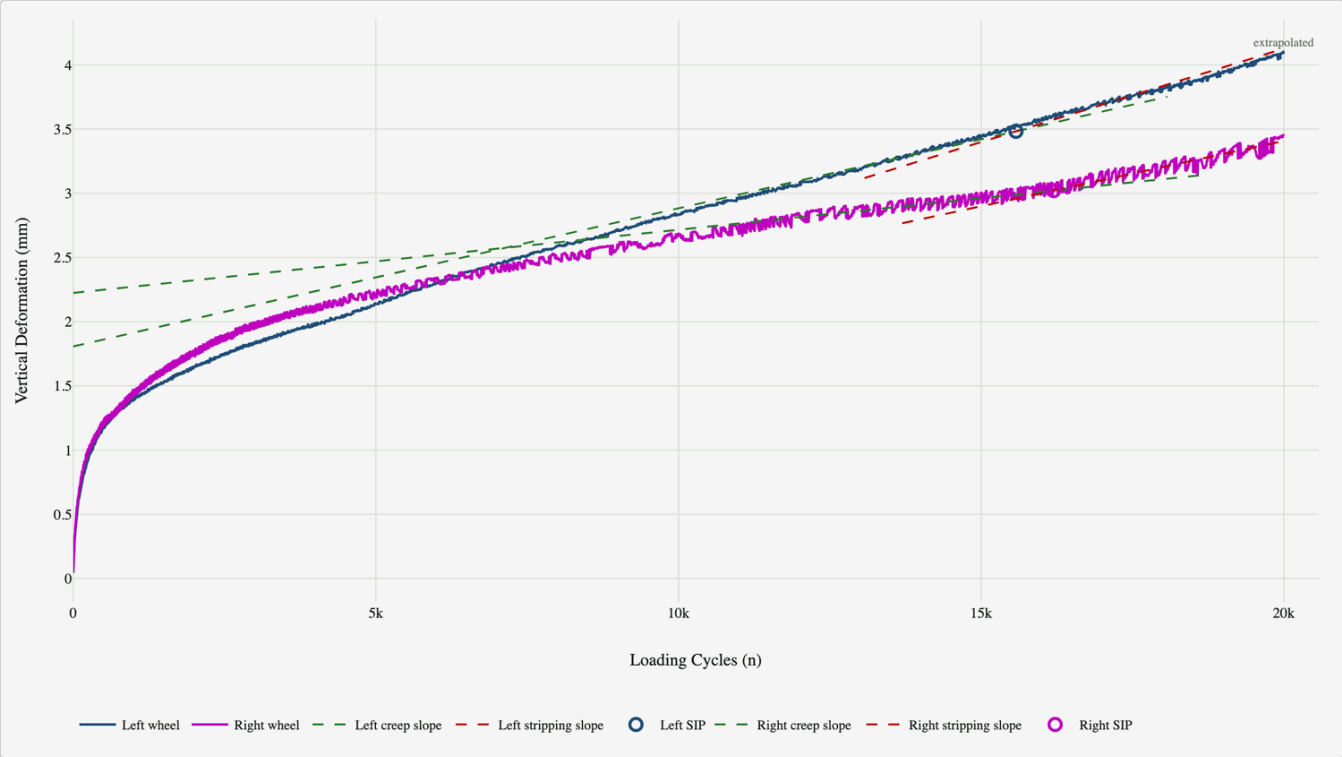


Figure 11.7 — *Both wheels with their creep and stripping slopes and the stripping inflection point.*

Checking the slopes

A 6th-order polynomial is fitted to each wheel; the stripping slope is taken at the steepest part of the second half of the record and the creep slope at the inflection before it. This is automatic, but the automatic choice is not always right — particularly on a noisy record, where it can latch onto the beginning of the curve.

Version 2 asked “Are the slopes okay?” for every wheel of every file. Version 3 draws them instead, so you can look. If a slope is wrong:

1. Switch the chart to **Selected + construction**.
2. Press **strip left** (or creep left, strip right, creep right).
3. Click the chart where that slope should be measured. You are narrowing the search, not setting the value by hand — the slope is still computed from the fitted polynomial there.

❹ The result updates immediately. **Use automatic slopes** reverts.

> **Two behaviours to be aware of, both carried over from Version 2.** Rutting is reported as the fitted polynomial evaluated at exactly 20 000 passes — if your test stopped earlier, that value is extrapolated, and the chart shades the region where this happens. When you set a slope by hand, the reported rut switches to the maximum measured depth instead.

> The stripping inflection point is only reported when the ratio of stripping slope to creep slope is 2 or greater. Below that the software leaves it blank, exactly as Version 2 did — the ratio is shown in the table so you can see why.

11.8 HT-IDT

ASTM D6931 (modified, high temperature)

High-temperature indirect tensile strength: a cylindrical specimen is loaded across its vertical diametral plane and the peak load at failure gives the IDT strength.

INPUT	Two columns: stroke and load.
TEMPLATE	Yes — two-channel. For the InstroTek Smart Loader export use **SmartJIG to RAAT**; for the raw MTS export use **MTS TWSX (stroke mm, load kN)**.
DIMENSIONS	**Thickness** and **diameter** in mm; air voids optional.
REPORTS	Peak load (N) and IDT strength (kPa).
PER REPORT	Many specimens.

This is the simplest module: load the files, enter thickness and diameter, run, export.

11.9 IDEAL-CT

ASTM D8225

An indirect tensile cracking test on a full-size gyratory specimen. Cracking resistance is expressed as the CT Index, derived from the fracture energy and the post-peak slope.

INPUT	Two columns: stroke and load.
TEMPLATE	Yes — two-channel. **Default – IDEAL-CT** expects positive stroke in mm from zero in column 1 and positive load in kN in column 2. For the raw MTS export use **MTS TWSX (stroke mm, load kN)**.
DIMENSIONS	**Thickness** and **diameter** in mm; air voids optional.
REPORTS	Peak load, L, tensile strength (kPa), fracture energy G_f (J/m^2), slope S, G_f/S, $G_f/S \cdot (L/D)^2$ and the CT Index.
PER REPORT	Many specimens.

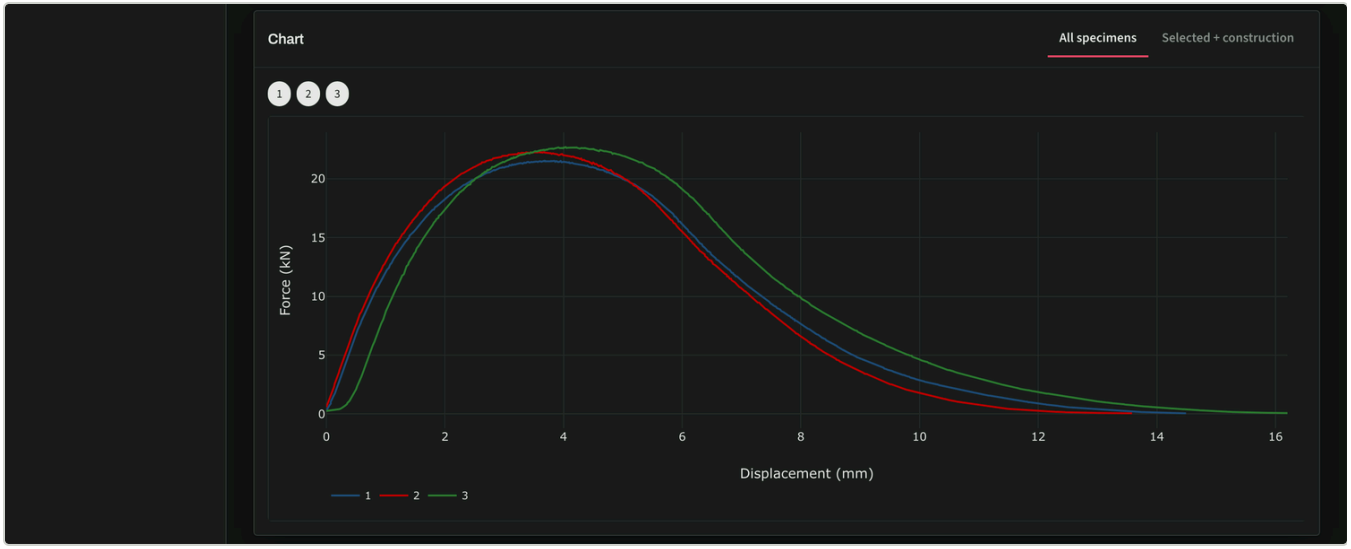


Figure 11.9 — *Three replicates overlaid. Switch to Selected + construction to see how the slope and L were derived for any one of them (Figure 7.1).*

A pass/fail criterion can be entered in the Export panel; it is printed beneath the results table on the report.

11.10 Intermediate Temperature SCB

AASHTO TP 124

A semi-circular specimen with an initiated crack is loaded at intermediate temperature. Resistance to further cracking is expressed as the Flexibility Index, derived from the fracture energy and the slope at the post-peak inflection point.

INPUT	Two columns: stroke and load.
TEMPLATE	Yes — two-channel. For the raw MTS export use **MTS TWSX (stroke m, load kN)** — note the stroke on these files is recorded in metres.
DIMENSIONS	**Thickness** and **ligament length** in mm; air voids optional.
REPORTS	Maximum load, fracture energy G_f (J/m^2), slope (kN/mm) and the Flexibility Index.
PER REPORT	Many specimens.

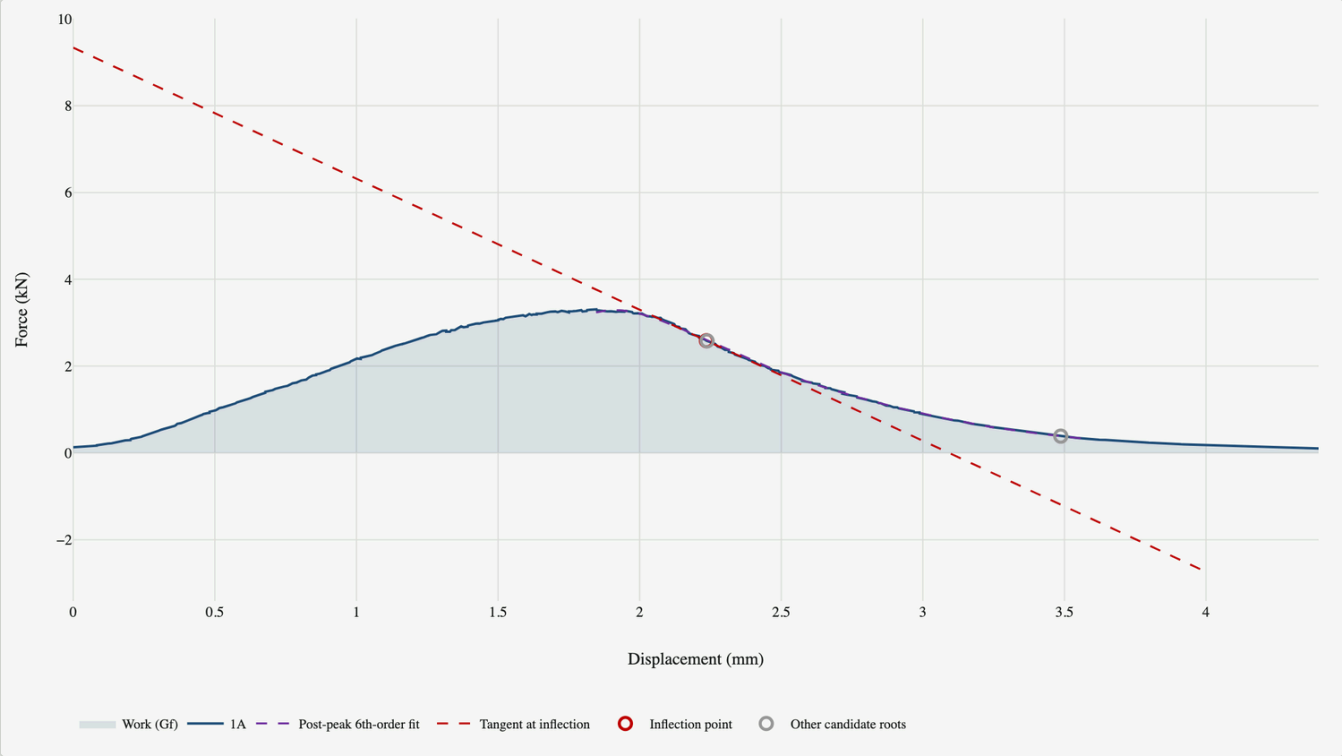


Figure 11.10 — *The post-peak 6th-order fit, the tangent at the inflection point, and the other candidate roots the search considered.*

Trimming

The end of every curve is trimmed automatically where the load falls to 0.1 kN — you do not need to do anything. If the start of the record contains a seating region, switch the chart to **Selected + construction**, press **Trim beginning** and click where the curve should start. **Undo trim** restores the full record.

The inflection point

The slope is measured at the post-peak inflection point, found by fitting a 6th-order polynomial and locating the root of its second derivative. Occasionally no valid root is found — the specimen is then marked **pick needed** in the results table and a note appears.

1. Select that specimen and switch to **Selected + construction**.

2. Press **Pick inflection point** and click the curve where the slope should be evaluated.
3. The Flexibility Index updates and the row is marked **manual pick**. **Use automatic** reverts.

> You can also override a slope the software found on its own if it looks wrong — the same check Version 2 performed with its "Is this slope ok?" prompt, except that here you can see the tangent before deciding. Save the session (section 10) so the pick is recorded.

SECTION 12

Troubleshooting

Files

SYMPTOM	CAUSE AND FIX
A file shows a red **error** badge after adding	Hover it for the message. Almost always the wrong template — the data-start row or the column numbers do not match. Open **File templates**, load that file as a sample and check the preview.
“No numeric data found”	The data-start row is past the end of the numbers, or the columns are empty. Check the sheet number too, if the workbook has more than one.
“Legacy binary .xls is not supported”	Re-save the file as `.xlsx` or `.csv`.
Curves look upside down or run backwards	The template's sign corrections are wrong. Displacement should increase from zero and load should be positive. Toggle **Negative** and **Offset from start** on the affected channel.
Values are out by a factor of 1000	Wrong units on a channel — usually stroke in metres read as millimetres, or load in newtons read as kilonewtons. The Intermediate SCB files from the MTS frames record stroke in *metres*.

Analysis

SYMPTOM	CAUSE AND FIX
“Missing required specimen dimension(s)”	A starred column is empty on that row. Use the ⇣ button to copy a value down a column.
A specimen is marked **pick needed** (SCB)	No inflection point was found automatically. Pick one on the chart — see 11.10.
A specimen is marked **no tail** (Cold SCB)	The tail extrapolation would not converge. Fracture energy is reported without the tail. Check the post-peak record is complete and decays smoothly.
“No creep-slope location found before the stripping location” (Hamburg)	The automatic search failed on a noisy record. Set both slopes by hand — see 11.7.
A row is marked **extrapolated** (Hamburg)	Not an error. The test stopped before 20 000 passes and the reported rut is extrapolated, as in Version 2. The shaded band on the chart shows how far.
A COV badge is amber	Also not an error — the scatter is above your warning threshold. Check the highlighted replicate in the statistics panel for a mis-typed dimension.

Dynamic Modulus

SYMPTOM	CAUSE AND FIX
“no ' Dynamic modulus (MPa)' row found”	That file is not an AMPT summary. Load the `…Sum.csv` files, not the raw data files.
“Calculate E*max from volumetrics first”	Step 1 has not been completed. Every master curve needs E^*_{max}.
“VTS shifting needs binder A/VTS”	Complete step 2, or run Arrhenius (EA) shifting instead, which does not need the binder.
“files have different frequency sets”	Informational. Frequencies present in some files but not others are dropped from the average.

Application

SYMPTOM	CAUSE AND FIX
The window opens blank or white	The WebView2 runtime is missing or out of date. Run Windows Update, or install the free WebView2 Evergreen runtime from Microsoft.
Export appears to do nothing	The file is still written to the application's `exports` folder — press **Open exports folder**.
A saved session will not reopen its files	The raw data was moved or deleted. Sessions reference the original file locations.

APPENDIX A

Demo data reference

The `Demo Data` folder that shipped with Version 2 works unchanged, and is the quickest way to confirm an installation behaves. Each folder holds the machine files together with the summary report Version 2 produced from them, so you can compare directly.

MODULE	FILES	TEMPLATE	DIMENSIONS
IDEAL-CT	Demo File 1–3	MTS TWSX (stroke mm, load kN)	t 62, D 150
Intermediate SCB	Demo File 1–4	MTS TWSX (stroke **m**, load kN)	t 50, ligament 58
Cold SCB	Demo File **4, 5, 6**	MTS TWSX 4-channel (stroke m, load N)	t 25, r 75, notch 15
DCT	Demo File **1, 3, 4**	MTS TWSX 4-channel (stroke mm, load kN)	D 150, t 50, ligament 83
HT-IDT	Demo File 1–3	SmartJIG to RAAT	t 150, D 150
Bond Strength	Demo File 1–3	MTS TWSX (stroke mm, load kN)	any
APA	APA, 2% AV BRIC	—	—
Hamburg	HAM Demo File 1	—	—
Beam Fatigue	Beam_RR_Berger_#7	Default – Beam	—
Dynamic Modulus	Winslow folder	—	target 4/20/45 °C, ref 20 °C

> The Cold SCB and DCT reports were not made from the first files in their folders — the bold numbering above is the set that matches the printed report. The dimensions listed are the ones printed on those reports.

APPENDIX B

Where the equations live

For whoever maintains the software. Skip this if you are only running tests.

Every calculation is in `backend/raatpack/equations/` — one file per test. Each file has its constants named at the top and the reported quantities computed in a single marked block, so a formula can be found and changed without reading the rest of the program:

```
SLOPE_WINDOW_UPPER_FRAC = 0.85   # post-peak slope window, upper bound
SLOPE_WINDOW_LOWER_FRAC = 0.65   # post-peak slope window, lower bound
DISPLACEMENT_FRAC       = 0.75   # L is read at this fraction of peak load
REFERENCE_THICKNESS_MM  = 62.0   # D8225 normalising thickness

# ---- reported quantities -------------------------------------------------
Gf       = U / (thickness_mm * diameter_mm) * MM2_TO_M2
CT_index = Gf_S * L_D * thickness_mm / REFERENCE_THICKNESS_MM
```

Adding a value to a report means adding it to that block, to the module's `RESULT_FIELDS` list, and to its result record — the tables, charts and every export format pick it up from there.

MATLAB-equivalent numerics — polynomial fitting and roots, bounded minimisation, root finding, the power and robust exponential fits, and Excel's AVERAGE/STDEV behaviour — are in `backend/raatpack/numerics/matlab.py`.

Regression tests

`cd backend && python -m pytest` runs every module's demo data through the analysis and compares the result against the numbers printed in the Version 2 reports. Change a formula and break a published value, and the test names which module and which quantity moved. Run it after any edit.

Version 2 behaviours kept on purpose

These affect reported values. They were reproduced deliberately so that results stay comparable with the laboratory's historical data. Changing any of them is an engineering decision, not a bug fix.

WHERE	BEHAVIOUR
Cold SCB tail	The power-law exponent is bounded *above* by −2 rather than fixed at −2 — the original code's second options call discarded the lower bound. Fixing it changes fracture energy by 3–10 %.
Hamburg rut	Reported as the fitted polynomial at exactly 20 000 passes even when the test ended earlier. The interface now shades the extrapolated region rather than hiding it.
Hamburg wheels	The last two data points of each wheel are dropped, a consequence of how the original indexed the record.
Cold SCB / DCT templates	The correction rows labelled Time, Stroke, COD and Load are applied to COD, Load, Time and Stroke respectively, and the unit rows are read in that same order.
Intermediate SCB	The inflection-point search window is measured over the whole curve, and the polynomial is not refitted after a manual pick.
Beam Fatigue	The AASHTO fit minimises absolute residuals; the exported ASTM trendline omits its constant term.
Dynamic Modulus	The activation energy is held at 2×10^5 by the fitting bounds, and target temperatures are rounded to whole °F before shifting.

Verification status

IDEAL-CT, Cold SCB, Hamburg, HT-IDT, DCT and APA reproduce the Version 2 reports to the precision those reports print. Intermediate SCB matches on three of four demo specimens and is within 0.4 % on the fourth. Beam Fatigue matches on the ASTM results and is within 0.03 % on the AASHTO $n_{f,50}$. The Dynamic Modulus master curve matches to 0.01 % when given the same E^*_{max}. Bond Strength has no Version 2 demo report and so has no verified value — its calculation was checked by hand against the original code.

APPENDIX C

Running from source

For development, and for running the software before the packaged application has been distributed. Works on Windows, macOS and Linux.

You need Python 3.8 or newer. Node.js is only needed if you intend to change the interface.

```
cd RAATPackv3/backend
python -m venv .venv
.venv\Scripts\activate          # Windows
source .venv/bin/activate       # macOS / Linux

pip install -e ".[dev]"
python -m pytest                # optional: the regression suite

python -m uvicorn raatpack.api.app:app --port 8765
```

Then open **http://127.0.0.1:8765** in a browser. The backend serves the already-built interface, so there is nothing else to start.

Building the Windows application

On a Windows machine with Python 3.11+ and Node 18+:

```
.\build\build_windows.ps1
```

This builds the interface, installs the analysis engine with pywebview and PyInstaller, and writes `build\dist\RAATPack\RAATPack.exe`. Distribute the whole `RAATPack` folder.

Changing the interface

```
cd RAATPackv3/frontend
npm install
npm run dev        # http://localhost:5173, proxies to the backend on 8765
```

Run `npm run build` when finished so the packaged application picks up the change.
